Report 3: The Inverse Zak Transform and the Level-0 Loopback

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S3

Mayank Mudgal (23B3918)

October 12, 2026

1 What this report is about

- The transmitter must turn the symbols on the delay–Doppler (DD) grid into an actual waveform. This is the **inverse Zak transform (IZak)**.
- The receiver does the reverse with the Zak transform of Report 2.
- The first sanity check of the whole chain is the **Level-0 loopback**: no channel, no noise, so what comes out must equal what went in.

2 The concept in brief

- **IZak recipe (two steps).**
  1. For every delay row m, take an N-point IFFT across the Doppler index n:
  $$v_m[q] = \frac{1}{N} \sum_{n=0}^{N-1} X[m,n]\, e^{j2\pi nq/N}.$$
  2. Put each value $v_m[q]$ on the time axis as a pulse anchored at sample $a = Dm + Pq$ (Report 1; the pulse's reference tap sits there, so the rectangular pulse used here starts at that sample) and add everything up:
  $$s[a] = \sum_{m,q} v_m[q]\; g[a - Dm - Pq].$$
- **What one DD symbol becomes.** A symbol $X[m,n]$ turns into a train of pulses at delay position m, one per period, whose phase rotates by $2\pi n/N$ from one period to the next. This is exactly the "spinning pulse train" of Report 2, which the Zak transform sends back to a single point. The IZak is just the sum of such trains, each weighted by its symbol.
- **The pulse g.** With D=1 the pulse is a single sample. With D>1 we use a rectangular pulse D samples wide and unit energy (height $1/\sqrt{D}$): each value is simply held for D samples.
- **Level-0 loopback.** $X \xrightarrow{\text{IZak}} s \xrightarrow{\text{Zak}} Y$ with nothing in between. The expectation is $Y = X$ exactly (up to rounding), for any D.
- **Why it works.** Different delay rows land on different samples and the pulses do not overlap, so the Zak transform can separate them again. (Pulses that overlap break this; that is the topic of Report 4.)
- **Scaling.** The IFFT has the factor $1/N$ and the Zak transform does not, so the waveform energy is $1/N$ of the energy on the DD grid, i.e. $E_s = E_X/N$.

One-line memory aid: IZak = IFFT along each row + place the values as pulses at $a = Dm+Pq$. Zak undoes it exactly when the pulses do not overlap.

3 What was simulated

- A full loopback on a random 16QAM grid (M=16, N=8, D=1): the symbols, the waveform and the recovered symbols.
- A sparse grid with only four occupied cells, to see where each delay row ends up in time (D=4).
- The D=4 waveform next to the D=1 waveform, to show the effect of the pulse.
- The phase trajectory of single Doppler bins, to see how Doppler becomes rotation.
- The loopback error and the energy for $D = 1, 2, 4, 8, 16$, and for 300 random frames.

4 Results and discussion

4.1 The loopback is exact

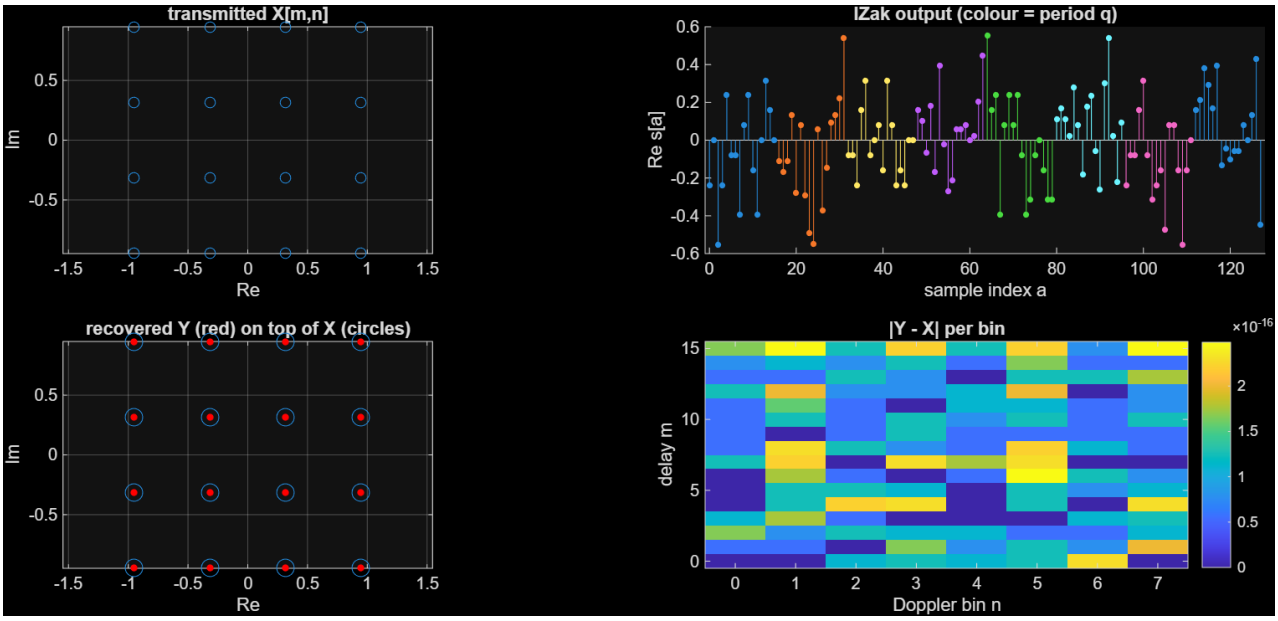


Figure 1: Random 16QAM grid. Top: transmitted symbols and the IZak waveform (real part, colour = period q). Bottom: recovered symbols (red) on the transmitted ones (circles), and the error per bin.

- **Expected:** $Y = X$ for every bin.
- **Observed:** the red dots sit exactly on the circles, and the error is at most 2.4×10^{-16}, i.e. rounding level.
- The waveform looks like noise, although it carries a clean 16QAM grid; this is a first hint that the DD grid is a convenient place to look at the signal.
- Takeaway: IZak followed by Zak gives the original symbols back.

4.2 Where do the delay rows go in time?

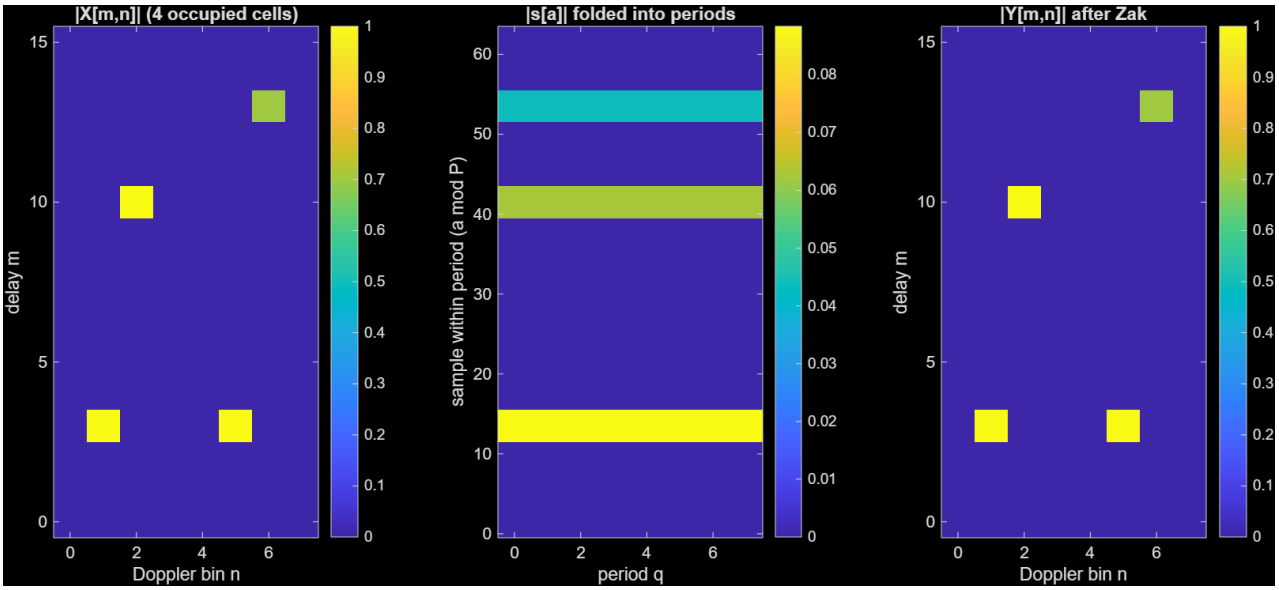


Figure 2: Four occupied cells (D=4). Left: $|X|$. Middle: $|s|$ folded into periods (vertical axis: sample within the period). Right: $|Y|$ after the Zak transform.

- **Expected:** each occupied delay row m appears as a band of D=4 samples starting at $4m$, in every period.
- **Observed:** bands at samples 12–15 (m=3), 40–43 (m=10) and 52–55 (m=13), identical in all 8 periods. The m=3 band is the brightest because two symbols share that row.
- The band heights follow the symbol sizes: a symbol of magnitude 1 gives $1/(N\sqrt{D}) = 0.0625$ (m=10) and a symbol of magnitude about 0.7 gives about 0.044 (m=13), as in the plot's colour scale.
- The right panel reproduces X exactly.
- Takeaway: the delay row sets *where* in the period the energy sits; the Doppler index of a single symbol only changes its phase from period to period, not its magnitude.

4.3 The effect of the pulse

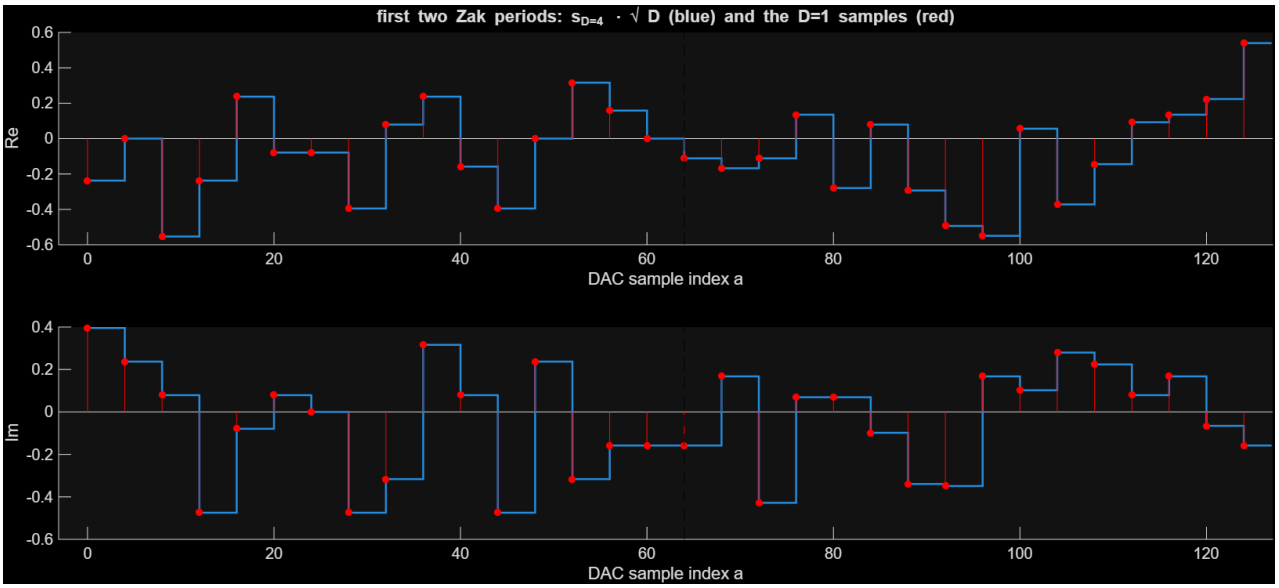


Figure 3: First two Zak periods: waveform for D=4 scaled by $\sqrt{D}$ (blue) and the D=1 samples (red).

- **Expected:** with a rectangular pulse the D=4 waveform just holds each D=1 value for 4 samples.
- **Observed:** the blue staircase passes through every red sample and is flat between them; the script asserts that the difference is below 10^{-12}.
- Takeaway: oversampling with a rectangular pulse adds no new information; it only draws each value wider (sample-and-hold).

4.4 Doppler is a rotation from period to period

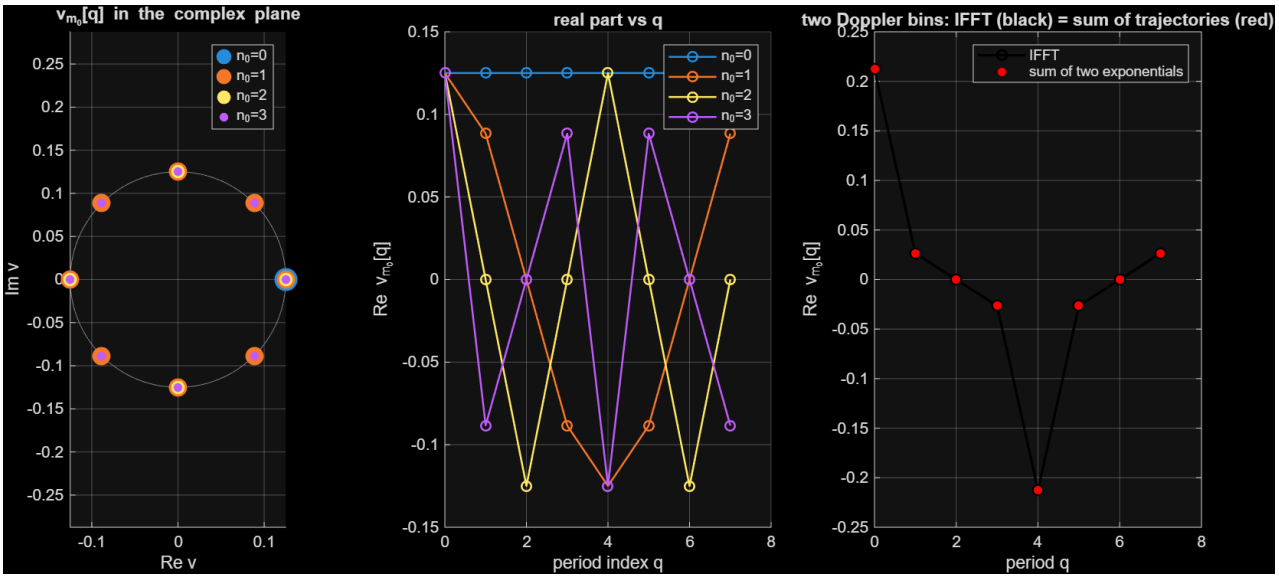


Figure 4: Left and middle: $v_{m_0}[q]$ for a single symbol in Doppler bin $n_0 = 0, 1, 2, 3$. Right: two Doppler bins together; the IFFT result (black) equals the sum of the two single-bin curves (red).

- **Expected:** bin n_0 rotates by $2\pi n_0/N$ per period; so n_0=0 stays put, n_0=1 visits the 8 points of a circle once, n_0=2 twice, n_0=3 three times. The circle radius is $1/N = 0.125$.

- **Observed:** all four curves lie on a circle of radius 0.125; n_0=0 is a constant, and the higher bins zigzag faster in the real-part plot.
- The right panel confirms linearity: two bins together equal the sum of their separate rotations.
- Takeaway: *Doppler is how fast the pulse's phase turns from one period to the next.*

4.5 The error is rounding at every D, and the energy law holds

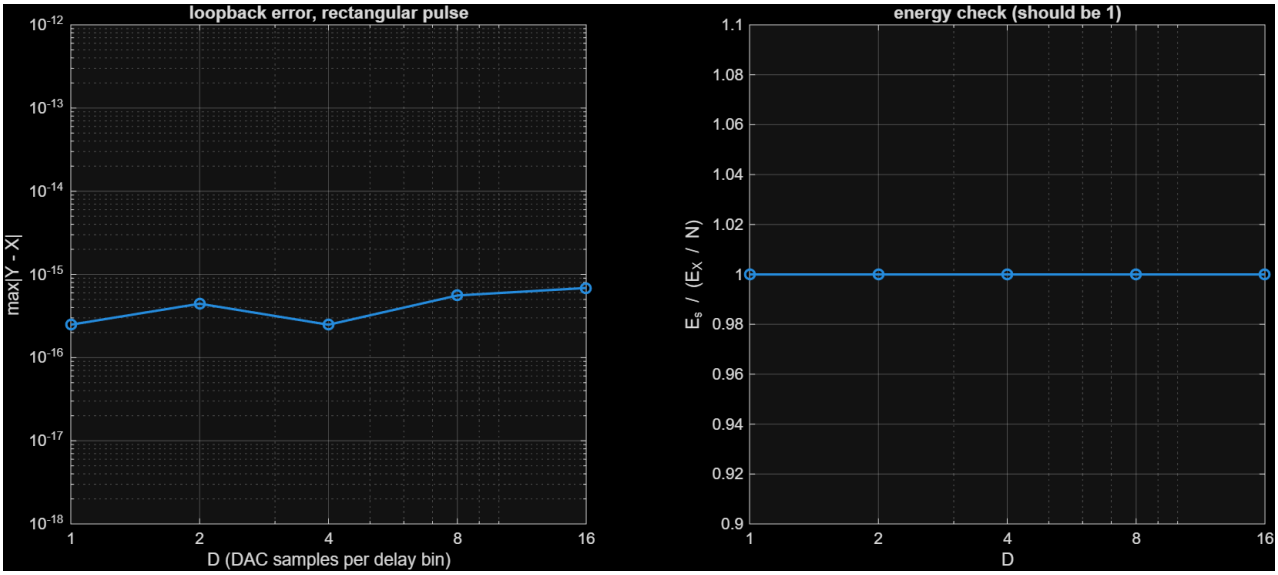


Figure 5: Left: loopback error against D (rectangular pulse). Right: $E_s/(E_X/N)$ against D.

Table 1: Numerical checks.

Check	Result
Loopback error, $D = 1, 2, 4, 8, 16$	2×10^{-16} to 7×10^{-16}
Loopback error, 300 random frames ($D = 1, 4, 8$)	2×10^{-16} to 10^{-15}
Fast IZak vs. literal triple sum	$\approx 5 \times 10^{-16}$
Fast IZak vs. explicit sample-and-hold	$\approx 10^{-18}$
$E_s/(E_X/N)$	1

- **Expected:** exactness for every D and the energy ratio of exactly 1.
- **Observed:** errors stay at the rounding level ($\sim 10^{-16}$) for all D, and the energy ratio is 1 for all D.
- In the 300-frame histogram of the script (not shown here) the D=1 and D=4 histograms coincide, and D=8 sits slightly higher ($\approx 10^{-15.2}$ instead of $\approx 10^{-15.5}$), still at rounding level. A plausible reason is that $1/\sqrt{4} = \frac{1}{2}$ is exact in binary, so D=4 adds no rounding beyond D=1; we confirmed in an independent NumPy run that the D=1 and D=4 errors are bit-identical and the D=8 ones are slightly larger.

5 Summary

- IZak = IFFT along each delay row, then place the values as pulses at $a = Dm + Pq$; Zak undoes it.
- For non-overlapping (rectangular, unit-energy) pulses the loopback is exact to $\sim 10^{-16}$ for every D tested.

- A delay row becomes a band of samples in each period; a Doppler bin becomes a rotation of the pulse phase from period to period.
- The waveform energy is $1/N$ of the DD-grid energy.

Appendix: details

- Figures come from the MATLAB script `S3_izak_loopback.m` (with local functions for the sampled IZak and Zak), using MATLAB's dark theme. The frame is treated as circular (the pulse of the last symbol wraps to the start); this does not matter for the non-overlapping rectangular pulse.
- The $E_s = E_X/N$ law is specific to this normalisation (IFFT with $1/N$, Zak without). The numbers in the table were reproduced independently in a second implementation (Python/NumPy).
- The exactness here depends on the pulse: it is true for a rectangular, unit-energy pulse D samples wide. Other pulses are studied in Report 4.